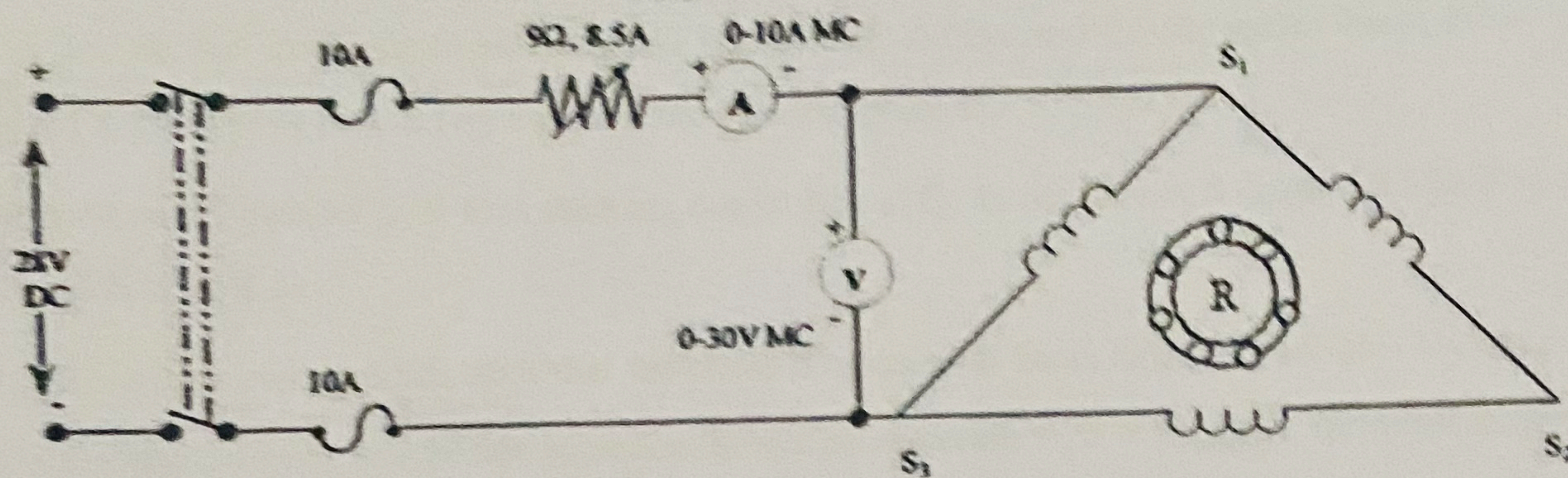




Procedure to draw the circle diagram : (Do not write in fair record)

1. Draw the lines by taking the current (I) in X-axis, voltage (V) in Y-axis. (V & I are line values).
2. From the No-load test find out the current I_0 and draw the OA vector with the magnitude of I_0 from the origin by suitable current scale, which lags the voltage (Y - axis) V by an angle ϕ_0 , where $\phi_0 = \cos^{-1} \left(\frac{W_{oc}}{\sqrt{3}V_{oc}I_{oc}} \right)$
3. From the current I_{sc} find out the I_{SN} (short circuit current corresponding to the normal voltage) through the formula $I_{SN} = I_{sc} \left(\frac{V_{rated}}{V_{sc}} \right)$, draw the OB vector with the magnitude of I_{SN} from the origin by the same current scale, which lags the voltage (Y-axis) V by an angle ϕ_{sc} where $\phi_{sc} = \cos^{-1} \left(\frac{W_{sc}}{\sqrt{3}V_{sc}I_{sc}} \right)$
4. Join the points B and A to get the output line AB.
5. Draw the parallel line for the X-axis from point A and for the Y-axis from point B upto the X-axis (point E), let both the lines intersect at point D.
6. Then draw the perpendicular bisector for the output line and extend it to the line AD. Let the point of intersection be C.
7. By keeping the point C as center and CA as radius draw a semi circle.
8. Let EB be the line of total loss [EB = ED + DB where ED = constant loss and DB = variable loss]
9. In the line DB locate the point G to separate the stator and rotor copper losses by using the formula, $\frac{\text{Rotor Copper Loss}}{\text{Stator Copper Loss}} = \frac{I_2'^2 R_2'}{I_2'^2 R_1} = \frac{R_2'}{R_1}$ where R_1 = stator resistance per phase and R_2' = equivalent rotor resistance per phase.

$$\text{Or } \frac{BG}{BD} = \frac{R_2'}{R_{01}} = \frac{\text{Rotor Copper Loss}}{\text{Stator Copper Loss} + \text{Rotor Copper loss}}$$

10. To get the torque line, join the points A and G